

Practice problems #2

1. True or False. Justify.

- (a) If $Ax = \lambda x$ for some scalar λ , then x is an eigenvector of A .
- (b) The eigenvalues of a matrix are on its main diagonal.
- (c) To find the eigenvalues of A , reduce A to echelon form.
- (d) For a matrix $A \in \mathbf{R}^{n \times n}$, A and A^T have the same set of eigenvalues.

2. Find the characteristic polynomial:

$$\begin{bmatrix} 6 & -2 & 0 \\ -2 & 9 & 0 \\ 5 & 8 & 3 \end{bmatrix}.$$

Answers or Hints on page 2.

#1: (a) F, (b) F, (c) F (justify them!)

(d) T (Hint: Since $\det M = \det M^T$, $\det(A - \lambda I) = \det (A - \lambda I)^T$. What is $(M + N)^T$?)

#2: $-\lambda^3 + 18\lambda^2 - 95\lambda + 150$